

Roll No. ....

**TPE-719**

**B. TECH. (PETROLEUM ENGINEERING)  
(SEVENTH SEMESTER)**

**MID SEMESTER EXAMINATION, Oct., 2022**

**CITY GAS DISTRIBUTION**

**Time : 1½ Hours**

**Maximum Marks : 50**

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each question carries 10 marks.

1. (a) For the following data given for a horizontal pipeline, base pressure ( $p_b$ ) through the pipeline given  $Z = 0.9188$  and viscosity  $0.0099$  cp.  
 $D = 13$  in,  $T = 80^\circ\text{F}$ ,  $L = 210$  mi,  $y_g = .70$ ,  $p_1 = 600$  psia,  $p_2 = 200$  psia,  $e = .0006$  in,  $T_b = 520^\circ\text{R}$ ,  $q = 1084025$  cfh (Show each step while answering).  
(CO1)

OR

- (b) A section of a pipeline system is to handle 200 MMSCFD gas flow. The pipeline inlet pressure is 900 psig and the flows to a dehydrator operating at 800 psig. The allowable working pressure of the pipeline is 1480 psi. The following data are given : STC 14.73 psi,  $60^\circ\text{F}$  , average

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temperature 80°F, compressibility 0.67, specific gravity of gas 0.85, viscosity 0.01 cp, length 5 miles. Determine D (Show each step while answering). (CO1)

2. (a) Using Panhandle A equation calculate the outlet pressure in a natural gas pipeline, with diameter 17.50 in. wall thickness 20 miles long. The gas flow rate is 150 MMSCFD at a flowing temperature of 70°F. The inlet pressure is 1000 psig and the gas gravity and viscosity are 0.6 and 0.00008 lb/ft-sec, respectively. Assume base pressure = 14.7 psia and base temperature = 60 OF. Assume that the compressibility factor  $Z = 0.85$  throughout and the pipeline efficiency is 0.95 (Show each step while answering). (CO1)

OR

- (b) Using Weymouth equation calculate the natural gas flow for gas and pipeline with the given information : (CO1)

$T_b = 60^\circ\text{F}$ ,  $P_b = 14.7$  psia,  $D = 12$  in, Pipe Length = .0947 mi, gas gravity = .65,  $T = 80^\circ\text{F}$ ,  $z = .919$ ,  $p_1 = 510$  psia,  $p_2 = 490$  psia (Show each step while answering).

3. (a) For the following data given for a horizontal pipeline, predict base pressure :  $D = 12.09$  in,  $L = 200$  mi,  $e = 0.0006$  in,  $T = 80^\circ\text{F}$ ,  $y_g = 0.70$ ,  $T_b = 520^\circ\text{R}$ ,  $p_1 = 600$  psia,  $p_2 = 200$  psia,  $z = .9188$ ,  $q_h = 1076035$  cfh (Show each step while answering). (CO1)

(3)

OR

- (b) Using Weymouth equation calculate base temperature for the following data given,  $P_b = 14.7$  psia,  $D = 10$  in, Pipe Length = 0896 mi, gas gravity = .61,  $T = 75^\circ\text{F}$ ,  $z = .919$ ,  $p_1 = 470$  psia,  $p_2 = 410$  psia,  $q_h = 272$  MMSCFD (Show each step while answering). (CO1)
4. (a) Define PNG and also describe PNG as a clean household energy, Industrial PNG and Commercial PNG. Write *five* advantages and disadvantages of PNG. (CO1, CO2)

OR

- (b) Write and explain three main operations of City Gas Station. (CO1, CO2)
5. (a) With the help of Typical Composition of natural gas in percentage table describe natural gas composition with compound having minimum and maximum percentage and also explain natural gas properties. (CO1, CO2)

OR

- (b) Explain District Regulating Station. Write at least five Advantages and disadvantages each of CNG, PNG and LPG. (CO1, CO2)